CAYHIC 2026 · FROM GENE TO MEDICINE

Free tools and resources

Everything from the talk, and more to explore. All free to use in a web browser. · Ulugbek Baymuradov · github.com/Bek/gene-to-medicine

Follow HER2 yourself the tools from the talk, with the clicks

Cancer Observatory	How common is each cancer where you live? → Fact sheets → your country
cBioPortal	Real tumor DNA data. Breast Invasive Carcinoma (TCGA, PanCancer Atlas) → gene ERBB2 → Submit. Then try "Cancer Types Summary".
UniProt	The protein's ID card: search "HER2 human" → P04626
AlphaFold DB	Predicted 3D shape of HER2. Blue = confident, orange = unsure.
RCSB PDB	1N8Z: the drug trastuzumab gripping HER2. Click "3D View" and spin it.
DepMap	Which cancer cells need which gene? Search ERBB2.
ClinicalTrials.gov	Every registered trial: Intervention "trastuzumab" → filter by country.
NHGRI costs	Why DNA data became cheap enough to share: cost per genome, 2001–2022.

Genes, genomes and cancer data

NCI Genomic Data Commons the full cancer genome datasets behind TCGA

OncoKB which cancer mutations have approved drugs

Human Protein Atlas where each protein is found in the body and in tumors

GTEx which genes are on in which healthy tissue

ENCODE map of the switches that control genes

UCSC Genome Browser · **Ensembl** browse the human genome

ClinVar is this DNA change linked to disease?

gnomAD how common a DNA change is in people worldwide

Drugs and targets

Open Targets evidence linking genes to diseases, and drugs to targets

ChEMBL millions of drug-like molecules and lab results

PubChem look up any chemical or drug

STRING which proteins work together

Reactome pathways: the cell's wiring diagrams

Protein structure and design

AlphaFold Server predict a protein's shape, and how it binds other molecules

ColabFold run structure prediction free in Google Colab

ESM Atlas 600 million+ predicted protein shapes

Mol* 3D viewer for any structure, in the browser

Foldseek find proteins with a similar shape

SAbDab every antibody structure solved so far

RFdiffusion AI that designs new proteins

Single cells and virtual cells

CELLxGENE explore millions of single cells, no code needed

Human Cell Atlas a map of every cell type in the body

Arc Virtual Cell Atlas data used to train virtual-cell models

scverse free Python tools for single-cell data, with tutorials

Virtual Cell Challenge 2025 results how the winners did it

Competitions you can enter

laptop + free cloud computers; check each site's rules

Virtual Cell Challenge	Predict how cells react when a gene is switched off. Top prize $100k (half cash). Age 18+. **Final entry 5 Nov 2026**
Virtual Embryo	Predict how embryo cells develop over time and space. $8k top prize per track; up to $200 for a helpful tutorial. Team captain 18+. **Final 2 Dec 2026**
Kaggle	Many prize contests, with free GPU notebooks. Search “Enveda CASMI”: identify molecules from lab measurements, $50k total. **Final 14 Dec 2026**
DREAM Challenges	Open biomedical data challenges, often with paper co-authorship.
CASP	The protein-structure prediction contest where AlphaFold made history.

Under 18?

International Olympiad in AI

International Biology Olympiad

Regeneron ISEF through your national science fair

Breakthrough Junior Challenge ages 13–18, a 2-minute science video

Virtual Embryo community award write a tutorial, no competing needed

Free computers and learning

Google Colab run Python with a free GPU in the browser

Kaggle Learn short free courses: Python, machine learning

Rosalind learn bioinformatics by solving problems

EMBL-EBI Training free courses on every database above

Galaxy analyze real DNA data with no code

Bioconductor R tools for genomics

3Blue1Brown how neural networks work, visually

Reading and careers

PubMed search published papers

bioRxiv new research before it is published

ISCB computational biology society, with a student council

US job outlook what jobs pay and how fast they grow

How to start tonight: 1) open the tools above on one gene you care about; 2) enter one competition, even with a simple first model; 3) write one page about what you found; 4) email a lab head or program coordinator with that page.

Free tools, no sponsorship. Links checked 28 September 2026. Slides and this list: **github.com/Bek/gene-to-medicine**